Network capacity is sometimes compute capacity: an accounting framework and decision boundaries for infrastructure investment in large-scale AI training

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Abstract

Operators of large accelerator fleets repeatedly choose between buying more accelerators and improving the infrastructure around the accelerators they already own. The metrics in common use do not support that comparison. Model FLOPs Utilization measures how well a step runs and is blind to failures; Effective Training Time Ratio measures availability and credits communication stall as productive work. Neither expresses an infrastructure improvement in units comparable to an accelerator purchase.

We introduce an accounting framework that assigns every accelerator-second to exactly one of four terminal fates, productive, blocked, discarded, or unavailable, and a substitution metric that expresses an infrastructure change as the number of additional accelerators delivering equal productive throughput. Calibrating a single compute-side parameter on one published training configuration, the model predicts a held-out configuration of the same system to within 1.6%, and reproduces both Daly's optimal checkpoint interval and a published closed form for expected training-time ratio.

Three findings follow. First, the gap between availability and useful capacity is large: at 16,384 accelerators our reference configuration reports an effective-training-time ratio of 0.889 while converting 0.627 of purchased accelerator-seconds into productive work. Second, the informal practice of converting avoided idle time directly into an equivalent accelerator count understates an intervention's value by a factor of 1.4 at 2,048 accelerators rising to 6.2 at 65,536, because a marginal accelerator is not fully productive. Third, no single intervention dominates: across the regimes we examine, four different interventions rank first, and under parameter uncertainty at 16,384 accelerators additional bandwidth ranks first in 0.6% of draws while halving the node failure rate ranks first in 52.3%. The title's claim holds in identifiable regimes and fails in others, and we report both. All results are analytical or simulated; none is a hardware measurement.

1 Introduction

An operator with a fixed budget can spend it on accelerators or on the infrastructure that keeps accelerators busy: more fabric bandwidth, a flatter topology, lower node failure rates, faster fault detection, faster restart, cheaper checkpoints. The first option is easy to price and easy to justify. The second is not, because the field lacks a metric that puts the two on the same axis.

The metrics we do have were built for other purposes. Model FLOPs Utilization (MFU) relates achieved model FLOPs to peak over the wall-clock of a running step; a job that loses a third of its life to restarts can still report a high MFU. Effective Training Time Ratio (ETTR), introduced by Kokolis et al. [2025], is productive runtime over the job's available wall-clock; it counts ordinary stepping as productive, so exposed collective communication is credited as useful work. The two overlap and have different denominators, so their product is not a capacity fraction.

This paper contributes an accounting framework, a substitution metric, an attribution rule, and a set of decision boundaries derived from them.

Contributions.

1. A four-fate decomposition of accelerator-seconds that is mutually exclusive and collectively exhaustive by construction, and is asserted numerically on every model evaluation (Section 4).
2. *Substitution-Equivalent Accelerators* (SEA), which expresses an infrastructure change as the accelerator count delivering equal productive throughput, and a measurement of how badly the informal alternative errs (Sections 4 and 6).
3. An order-independent attribution rule for interacting interventions, with the finding that recovery improvements are complements rather than substitutes (Section 6).
4. Decision-boundary maps showing which intervention is the strongest marginal investment in each regime, including the regimes where network investment is worth nothing and the regimes where no accelerator purchase can substitute for infrastructure quality at all (Section 6).

We do not contribute a new simulator, a new communication model, or a new reliability model. Each of those exists, and we deliberately reuse standard formulations so that our predictions can be checked against independent tools and closed forms.

2 Motivation and the decision problem

The immediate motivation is a class of infrastructure argument that is made qualitatively and decided quantitatively. Vendor material describes pooling compute across power-constrained campuses, synchronized millisecond traffic bursts, tail-latency sensitivity, automated hang detection, and high-resolution telemetry [Google Cloud, 2025]. Each of these is plausible and none is accompanied by an independent measurement of how much accelerator capacity it recovers. We treat that material as motivation only.

The decision problem is concrete. Given a fleet, a workload, and a marginal budget, which purchase yields the most productive accelerator time? Answering it requires (i) knowing where accelerator time currently goes, (ii) a common unit, and (iii) knowing how the answer changes with the operating regime.

3 Related work

Simulation and performance modeling. ASTRA-sim [Won et al., 2026], SimAI [Alibaba Cloud, 2025], and Calculon [Isaev et al., 2023] model distributed training performance at fidelities ranging from closed-form analytical to packet-level. None models failures, recovery, or checkpointing, and none expresses results as a substitution against accelerator count. Svedas et al. [2025] survey this space and catalogue total-cost models separately from performance simulators, which is itself evidence that the two are not coupled.

Reliability modeling and measurement. Kokolis et al. [2025] report eleven months of operation across two clusters, define ETTR, and give a closed form for its expectation. ByteDance Seed [2025] report 97% ETTR on a 9,600-GPU three-month job. AIReSim [Pattabiraman et al., 2026] simulates failure, recovery, scheduling, and repair; its authors state explicitly that it models no network topology, communication time, or bandwidth, and that its parameters are hypothetical. These works measure or simulate reliability; none prices reliability against network or accelerator spending.

Network cost. Wang et al. [2023] show that LLM traffic is sparse enough that removing the spine layer preserves training performance while cutting network cost by 38–77%. That is the dual of our question: they minimize network cost at fixed performance, while we compare marginal network, reliability, and accelerator spending at fixed budget. Cao et al. [2024] demonstrate the mechanism we rely on, that communication contention converts directly into lost GPU utilization.

Counterfactual accounting. Lin et al. [2025] use what-if analysis over a production trace to attribute lost time to stragglers, establishing the methodological precedent for counterfactual capacity accounting. Their scope is stragglers within one organization's fleet.

White space. Reading these together, communication modeling and failure modeling are each well served but rarely in the same model, cost comparison appears only at fixed performance, and no inspected work expresses an improvement as an equivalent accelerator count. We found no prior public work that combines all four.

4 Framework and methodology

4.1 Four-fate accounting

For a pool of N accelerators over a window of T seconds, every one of the NT accelerator-seconds is assigned exactly one terminal fate: **productive** (computation present in the delivered model state), **blocked** (powered and allocated but not computing: exposed communication, synchronization wait, pipeline bubble, checkpoint stall, restart), **discarded** (computed and then thrown away: everything between the last durable checkpoint and failure detection), or **unavailable** (owned but not usable: down, spare, or stranded).

$$P + B + D + U = NT \tag{1}$$

Two conventions make this exact. Classification is by *terminal fate*, not activity: a second spent computing inside a window later discarded is discarded, not productive. And re-execution is counted once: after a restart, the original attempt is discarded and the repeat is ordinary productive and blocked time. Charging both is the most common error in informal goodput accounting. Equation 1 is asserted on every model evaluation to a relative tolerance of 10^{-9}.

The *Useful Capacity Fraction* is $\mathrm{UCF} = P/(NT)$: the share of what was paid for that became useful work. Its denominator is the pool, so spares and stranded capacity reduce it.

4.2 Substitution-Equivalent Accelerators

Let $\Pi(N, c)$ be productive accelerator-seconds per second for a pool of N under configuration c. For a baseline c_0 at size N_0 and an intervention yielding c_1:

$$\mathrm{SEA}(c_1; c_0, N_0) = \Delta N \quad \text{such that} \quad \Pi(N_0 + \Delta N, c_0) = \Pi(N_0, c_1) \tag{2}$$

SEA prices the alternative correctly, because a purchased accelerator inherits the fleet's inefficiency rather than arriving fully productive, and it accounts for the cost of growth, because Π is recomputed at the larger size where communication volume is higher and job mean-time-to-failure is shorter. Where $\Pi(N, c_0)$ never reaches the target, SEA is unbounded: the improvement cannot be bought with accelerators at any quantity.

SEA requires no prices. It *is* the break-even cost: an intervention is worth funding when it costs less than SEA fully-loaded accelerators. This is why we report no currency figures anywhere in this paper.

The informal alternative, dividing avoided idle and repeated accelerator-seconds by the measurement window, assumes a marginal accelerator is fully productive. We compute it alongside SEA so its error can be measured rather than asserted.

4.3 Attribution

Π is nonlinear in c, so SEA is not additive across interventions: relieving one bottleneck changes what the others are worth. We attribute contributions with the Shapley value [Shapley, 1953] over all 2^k subsets, which is affordable because the model is cheap, and we verify the efficiency axiom as a test.

4.4 Model

Compute time follows the standard transformer FLOP accounting [Narayanan et al., 2021]. Collective time uses an alpha-beta cost model applied tier by tier across a scale-up domain, a full-bisection pod, and an oversubscribed cross-pod layer, with ring costs at each tier. Rank layout is modeled explicitly, so a data-parallel collective whose members sit one per scale-up domain does not incorrectly receive scale-up bandwidth. Synchronization cost uses the Gaussian extreme-value approximation for the maximum of N draws.

Reliability is modeled twice, independently. An analytical path uses renewal expectations; an event-driven path walks the wall clock, places checkpoints, draws exponential failure times, and attributes each second as it goes. They share only the parameter definitions. Agreement between them defines the model's validity envelope (Section 5).

5 Validation

5.1 Invariants and closed forms

Beyond Equation 1, the model satisfies limiting cases (unbounded bandwidth drives exposed communication to zero; a zero failure rate drives discarded and restart time to zero; an ideal cluster reaches UCF > 0.999) and monotonicity in bandwidth, failure rate, and oversubscription.

Two closed forms derived independently of this model are recovered from it. Daly's optimal checkpoint interval [Daly, 2006] is recovered by brute-force search over the model's own predicted productive time, within 10% at checkpoint costs of 10, 30, and 120 seconds. The expected-ETTR

Quantity	Published	Model	Error	Verdict
Held-out throughput (TFLOP/s per accelerator)	400	406.3	1.6%	pass ($\leq$10%)
Held-out BF16 MFU	41%	41.1%	0.1%	pass
Interruptions in 54 days	419	526.6	25.7%	pass ($\leq$35%)
Job-level ETTR	>90%	88.8%	1.2 pts	pass ($\geq$85%)
Job MTTF at 16,384 accelerators	1.8 h	1.803 h	0.2%	pass
Job MTTF at 131,072 accelerators	0.23 h	0.225 h	2.2%	pass

Table 1: External validation. The held-out configuration doubles data parallelism at fixed global batch, halving compute per step while spreading the gradient all-reduce across twice as many pods. Thresholds were fixed before the comparison ran.

expression of Kokolis et al. [2025] is matched within 5% at 2,048, 8,192, and 16,384 accelerators in the small-loss regime where that approximation applies.

5.2 External validation

We calibrate one compute-side parameter, kernel efficiency, on a single published anchor and then hold it fixed to predict a different configuration of the same system [Grattafiori et al., 2024]. No network or reliability parameter is tuned at any point.

The interruption threshold is deliberately loose because the failure rate comes from a different cluster than the one predicted. The rate implied by the published interruption count, 3.79×10^{-3} per node-day, sits inside the range reported across the two clusters of Kokolis et al. [2025] (2.34 to 6.50×10^{-3}).

5.3 A validation failure

The first run failed the ETTR check, predicting 0.600 against a published value above 0.90. The cause was two misspecified inputs, not the model: the configuration assumed no replacement capacity, so every failure waited a full physical repair, and assumed 60-second fully blocking checkpoint writes. Both contradict the source, which states the cluster held 24,000 GPUs while the job used up to 16,000 and that failures were handled by automation. An independent arithmetic check confirms the diagnosis: at the Daly optimum, combined checkpoint and lost-work overhead is approximately $\sqrt{2w/M}$, which for a 60-second blocking write and a 2.34-hour MTTF is about 12%, making an ETTR above 90% unreachable at this scale regardless of any other parameter. We report the failure rather than only the corrected result.

5.4 Validity envelope

The analytical and event-driven paths agree closely over most of the parameter space and diverge where mean-time-to-failure approaches recovery time. We define *recovery pressure* as the share of runtime consumed by discarded work plus restart. Below 0.25 the two paths agree within 1.4% on UCF; above it they diverge to 29.6% (Figure 1). Rows above the threshold are flagged in the raw output and excluded from headline claims; where such a case is load-bearing we recompute it with the event-driven path.

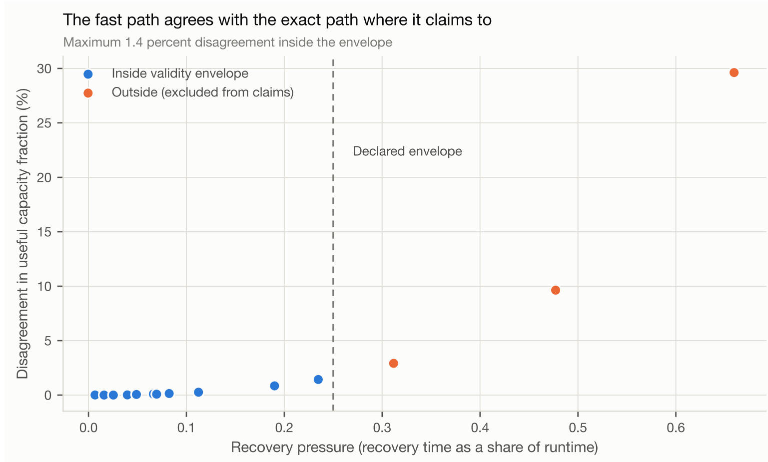

Figure 1: Cross-fidelity agreement. The fast path is trusted only where it claims to be valid, and the threshold is set from this data rather than assumed.

6 Results

All results below use a study configuration derived from the validated one and documented separately, so that no study parameter contaminates a validation claim.

6.1 Availability is not capacity

At 16,384 accelerators the reference configuration reports ETTR = 0.889 and UCF = 0.627, a gap of 26.1 percentage points. An operator reading only ETTR would conclude the fleet is 89% effective while 37% of purchased accelerator-seconds produced nothing. The gap is not a modeling artifact: it is what ETTR is defined to exclude, namely exposed communication and the capacity a job never held.

Figure 2 also refines the common framing. Communication is a roughly constant share of the loss across scale in this configuration; what grows is pipeline bubble and failure-related time. This is the first place the paper's title requires qualification.

6.2 The marginal accelerator

At the reference baseline the marginal accelerator contributes 0.502 productive accelerator-seconds per accelerator-second purchased. This single number drives the substitution metric: an intervention worth 141 productive accelerators is worth 281 purchased ones. Notably the four oversubscription curves in Figure 3 nearly coincide, meaning that at this configuration the spine is not what limits scaling, a second qualification of the title.

6.3 The informal metric is biased, and the bias grows

The informal metric equals 0.73 of SEA at 2,048 accelerators and 0.16 at 65,536, understating intervention value by factors of 1.4 and 6.2 respectively. The bias is structural: the informal metric implicitly divides by a marginal productivity of 1, and the true marginal productivity falls with scale. It errs most severely exactly where the investment decision is largest.

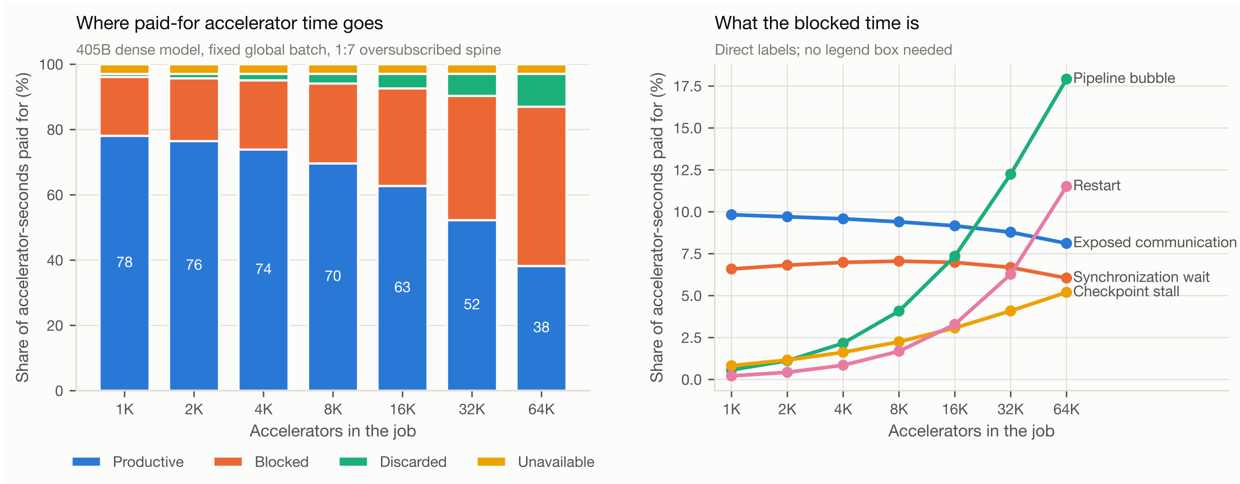

Figure 2: Where paid-for accelerator time goes as job size grows at fixed global batch. The productive share falls from 0.781 at 1,024 accelerators to 0.382 at 65,536. Exposed communication stays near 8–10% of the total throughout, while pipeline bubble and restart grow from roughly 1% combined to roughly 29%.

6.4 Interventions interact, and not always as substitutes

Excluding cases where SEA is unbounded, bandwidth and topology improvements are substitutes as expected (mean additivity error +6.5%), and a mixed network and reliability bundle likewise (+7.5%). Recovery improvements are *complements* (−8.9%): faster checkpointing shortens the optimal checkpoint interval, which shrinks the window of discarded work, which raises the value of faster detection and restart. This refutes the direction we hypothesized and is reported as such.

6.5 Break-even costs

Figure 5 gives the break-even costs directly. Two features matter for planning. Enlarging the scale-up domain and adding spares are worth nothing in every regime shown, because neither relieves a binding constraint at this configuration. And the spread across regimes is wide: faster restart ranges from roughly 280 to 2,000 accelerators depending on the regime, so a single quoted value for such an intervention would be misleading.

6.6 Decision boundaries

Figure 6 is the central result. Across 96 in-envelope cells, four different interventions rank first. At 16,384 accelerators the map divides between straggler control at lower failure rates and halving the failure rate at higher ones; additional bandwidth never wins. At 65,536 accelerators the picture changes: quadrupled bandwidth wins in the low-failure, heavily oversubscribed corner, faster restart wins a middle band, and the two 16K-scale winners retain the remainder.

Point rankings are not decisions, so we recompute the ranking over draws from documented parameter ranges (Figure 7). At 16,384 accelerators, halving the failure rate ranks first in 52.3% of draws, straggler control in 24.8%, faster checkpointing in 19.2%, faster restart in 3.1%, and quadrupled bandwidth in 0.6%.

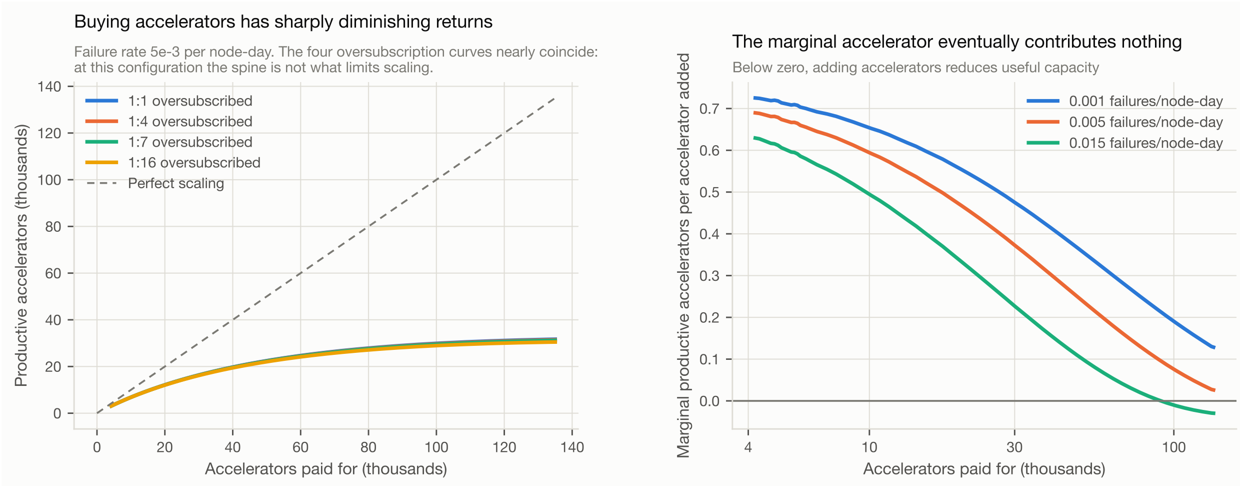

Figure 3: Productive throughput against pool size, and its derivative. At fixed global batch the marginal accelerator's contribution falls toward zero and, at high failure rates, below it.

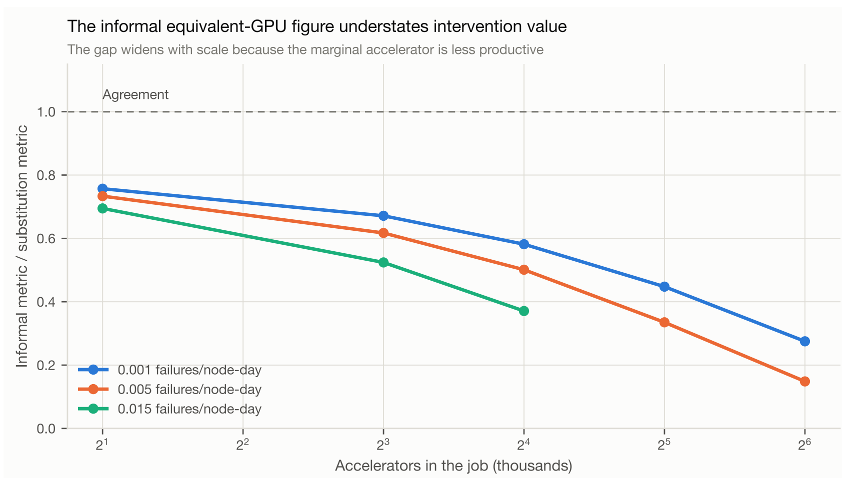

Figure 4: The informal equivalent-accelerator figure as a fraction of the substitution value.

7 Sensitivity and counterexamples

We sought negative cases deliberately.

Where network investment is worth nothing. With an already fast and flat network (3.2 Tb/s per accelerator, no oversubscription), every bandwidth and topology intervention has SEA $= 0$ exactly, while halving the failure rate is worth 954 accelerators. The mirror case also holds: with near-perfect reliability, halving the failure rate is worth zero while doubling bandwidth is worth 255.

Where nothing much matters. At 1,024 accelerators no intervention is worth more than 39 accelerators, because UCF is already 0.781. Small jobs do not justify infrastructure investment on capacity grounds.

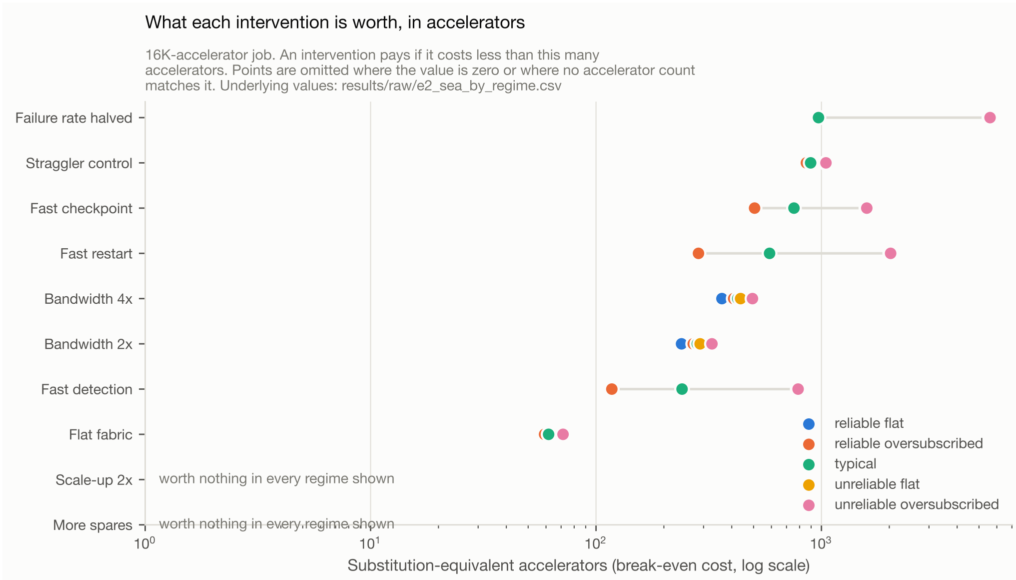

Figure 5: Substitution value of each intervention across five regimes at 16,384 accelerators, on a logarithmic axis. Because SEA is a break-even cost, each point states how much an intervention may cost, in fully-loaded accelerators, before it stops paying. Points are omitted where the value is zero or where no accelerator count matches the intervention.

Where communication dominates. Shortening the sequence to 2,048 tokens or the global batch to 512 sequences raises the value of doubled bandwidth to 871 and 882 accelerators respectively, because communication becomes a larger share of a smaller step.

Where accelerators cannot substitute at all. At approximately 131,000 accelerators under this configuration, productive throughput *decreases* as the pool grows: the event-driven path gives 29,682 productive accelerators at the baseline pool, 28,234 at 1.5×, and 25,016 at 2×, while halving the failure rate yields 38,075. No purchase matches the intervention. This case lies outside the fast path's validity envelope and was confirmed with the event-driven path.

8 Strategic implications

The results support three statements an infrastructure planner can act on.

First, measure capacity against what was purchased. A fleet reporting 89% effective training time may be converting 63% of its purchased accelerator-seconds into useful work, and the difference is where the budget argument lives.

Second, express improvements as break-even accelerator counts rather than as recovered time. The conversion is not a formality: the informal version understates value by up to an order of magnitude at large scale, and it cannot express the regime where no purchase substitutes.

Third, do not carry a single ranking across regimes. The best marginal investment at 16,384 accelerators with reliable nodes is not the best at 65,536 with an oversubscribed spine, and our own thesis fails at the more common operating points: additional bandwidth ranks first in under 1% of uncertainty draws at 16K scale.

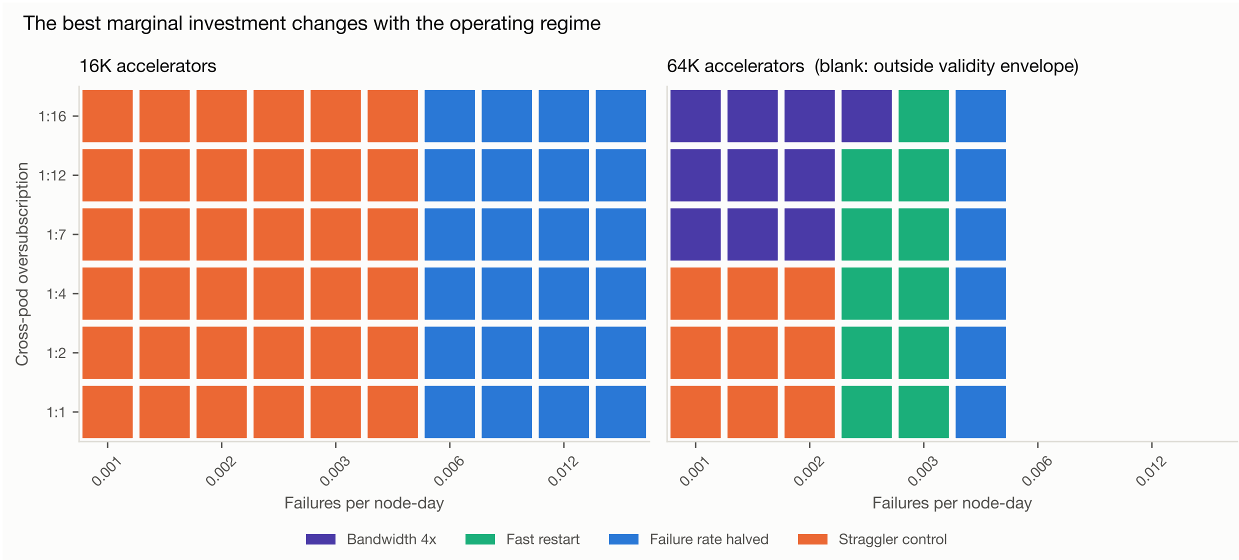

Figure 6: The intervention with the highest substitution value, over a failure-rate by oversubscription grid. Four different interventions rank first, and the winning set differs between the two scales. Blank cells fall outside the validity envelope.

9 Limitations and threats to validity

No hardware measurement. Every number here is analytical or simulated. The external validation compares against published measurements of a system we did not operate, using a configuration reconstructed from a paper.

The failure process. Interruptions are modeled as a per-node Poisson process. This reproduces reported job MTTF at 16,384 and 131,072 accelerators within a few percent, but misses the measured 1,024-GPU MTTF by a factor of 3.7. The source's own figures are not mutually consistent under inverse scaling, indicating that small-job interruptions are dominated by causes we do not model. Results below roughly 4,096 accelerators are indicative only. We also assume failures are independent; correlated failures would worsen outcomes and would raise the value of fault-isolation interventions we do not represent.

Scaling policy. SEA depends on how parallelism is re-planned as the pool grows. We assume strong scaling at fixed global batch, which is the policy for finishing a given run sooner. Weak scaling changes the magnitudes, though not the existence of rank reversals.

Placement. We assume dense, regular rank placement, the best case for the network. Fragmented placement would make every network intervention look better, so this assumption is conservative with respect to our own thesis.

Unvalidated counterfactuals. No published source reports a system measured before and after a network or reliability change, so the substitution metric's inputs are validated but the metric's output is not.

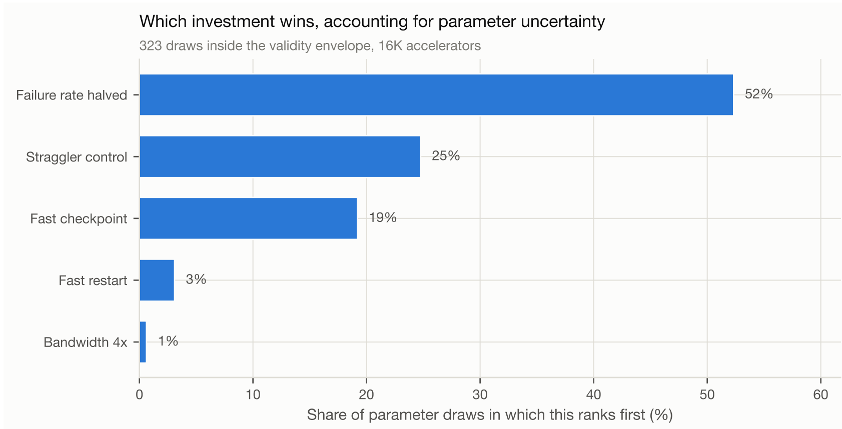

Figure 7: Share of parameter draws in which each intervention ranks first, over 323 in-envelope draws from documented parameter ranges at 16,384 accelerators.

Scope. Dense transformer pre-training, synchronous, single campus. No inference, no mixture-of-experts routing, no asynchronous training, no cross-campus pooling, no absolute costs.

10 Conclusion

Network capacity is compute capacity in identifiable regimes, and is not in others. Deciding which regime a fleet is in requires accounting that closes, a metric that prices the alternative correctly, and an honest treatment of the parameters that are not known. We provide all three, along with the cases where our own thesis fails.

The framework's most useful property may be its least dramatic one: because the substitution metric is a break-even cost, an operator can apply it without agreeing with any of our cost assumptions, of which we make none.

Reproducibility

The complete implementation, configurations, raw results, and figure scripts are available at `https://github.com/dimaggi-ai/network-vs-more-gpus`. The full experiment program runs in about ten seconds on a laptop and requires no accelerator access. `make reproduce` regenerates every number and figure in this paper; `make smoke-test` runs a fast subset. Raw results are immutable and carry provenance sidecars recording the model version, git commit, seed, and configuration digest. Every threshold in Section 5 was fixed before the corresponding comparison ran.

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